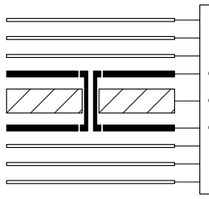
SN2 BREAKOUT BOARD FABRICATION DOCUMENT

LAYER STACKUP LEGEND



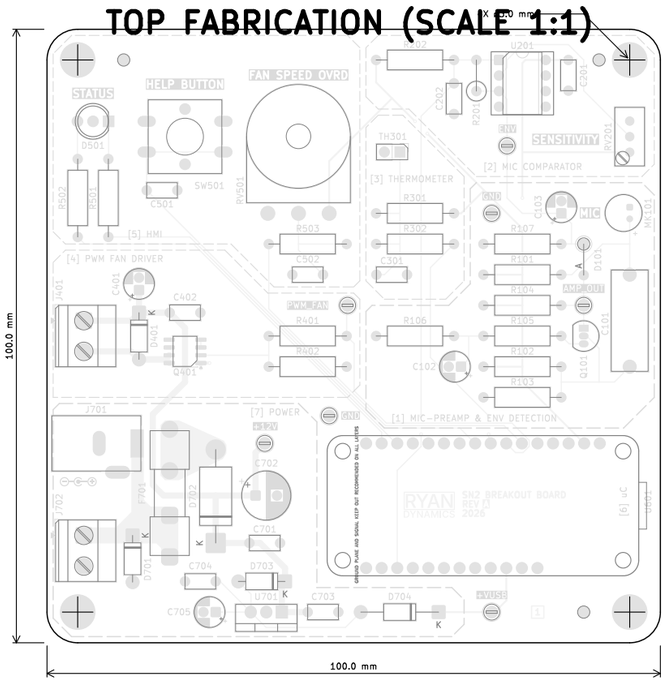
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig. PWR)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
Core		1.49mm	FR4_7628	Dielectric	
Copper	**L2 (Sig. GND)**	**0.035mm (1.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.6mm
Note: external layer thicknesses are specified after plating

IMPEDANCE TABLE

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0.2032	0.28	L2

TOP FABRICATION (SCALE 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.
2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
4) SURFACE FINISH: HAL SNPB
5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.
8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY RYAN DYNAMICS.
10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

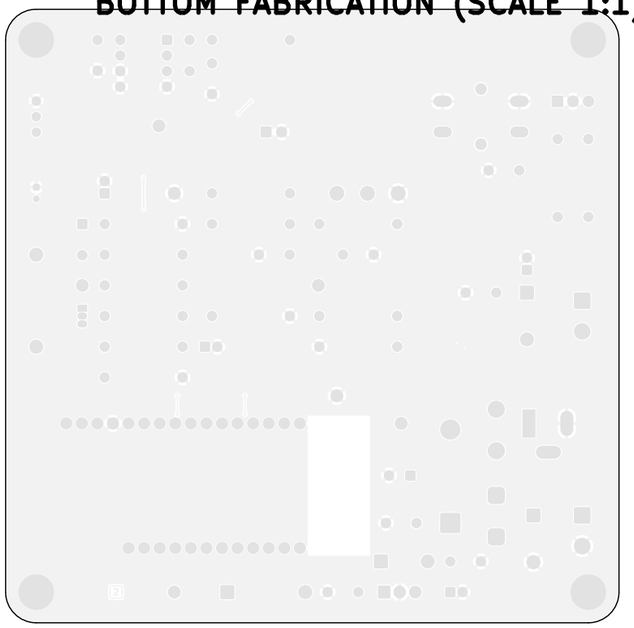
BOARD SIZE	100.000 × 100.000 mm
BOARD THICKNESS	1.600 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.200 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	1.800 mm
ANNULAR RING	0.150 mm
COPPER TO HOLE	0.254 mm
COPPER TO EDGE	0.250 mm
HOLE TO HOLE	0.254 mm

ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

PROJECT ELEC4740	Ryan Dynamics		
DRN 2024-04-13 R. HICKS	THIS DESIGN AND/OR DRAWING IS THE PROPERTY OF RYAN DYNAMICS AND SHALL NOTBE REPRODUCED WITHOUT AUTHORISATION.		
CHK			
ENG APP	SN2 BREAKOUT BOARD		
MFR APP			
SHEET PATH	GIT HASH 9d7cd56	DRAWING No FABRICATION	A3
FILENAME Sensor-Node-2-Board.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased) SHEET 1 OF 8	

SN2 BREAKOUT BOARD FABRICATION DOCUMENT

BOTTOM FABRICATION (SCALE 1:1)



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CHK		
ENG APP	SN2 BREAKOUT BOARD	
MFR APP		
SHEET PATH	GIT HASH 9d7cd56	DRAWING No FABRICATION
FILENAME Sensor-Node-2-Board.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased) SHEET 2 OF 8

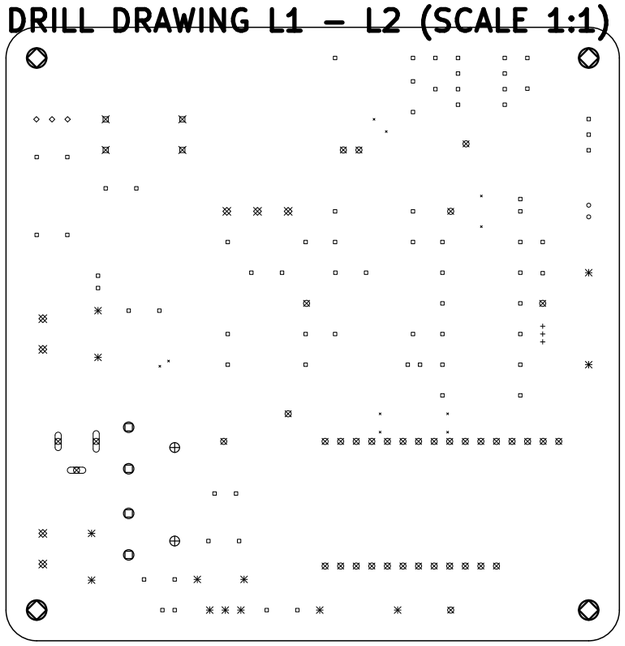
A3

SN2 BREAKOUT BOARD FABRICATION DOCUMENT

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	10	0,30mm (11,81mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Via
○	2	0,65mm (25,59mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
+	3	0,75mm (29,53mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
□	73	0,80mm (31,50mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
◇	3	0,90mm (35,43mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊠	37 + 3	1,00mm (39,37mils)	PTH	Round + Slot	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
✳	13	1,10mm (43,31mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊠	4	1,20mm (47,24mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊠	7	1,30mm (51,18mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
⊕	2	1,60mm (62,99mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
○	4	1,70mm (66,93mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
○	4	3,20mm (125,98mils)	PTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Pad
	Total 165					

DRILL DRAWING L1 – L2 (SCALE 1:1)



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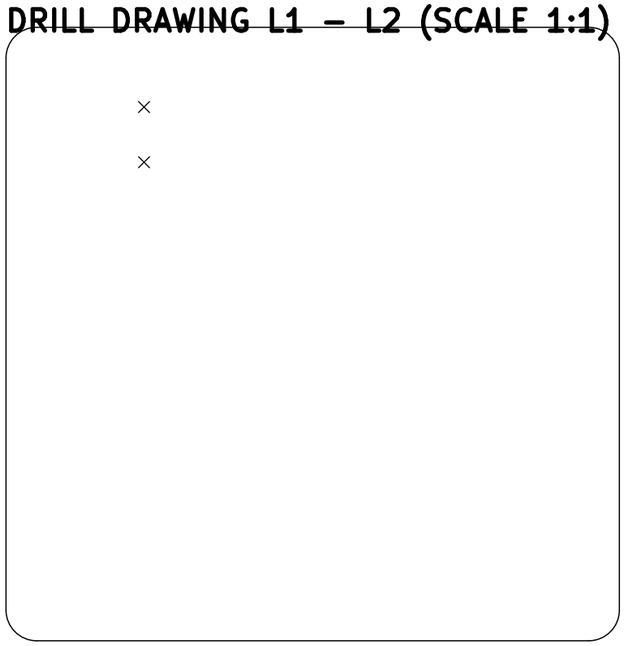
DRN 2024-04-13 R. HICKS			
CHK			
ENG APP	SN2 BREAKOUT BOARD		
MFR APP			
SHEET PATH	GIT HASH 9d7cd56	DRAWING No FABRICATION	A3
FILENAME Sensor-Node-2-Board.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased)	SHEET 3 OF 8

SN2 BREAKOUT BOARD FABRICATION DOCUMENT

DRILL TABLE

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	2	1.80mm (70.87mils)	NPTH	Round	L1 (Sig, PWR) - L2 (Sig, GND)	Mechanical
	Total 2					

DRILL DRAWING L1 – L2 (SCALE 1:1)



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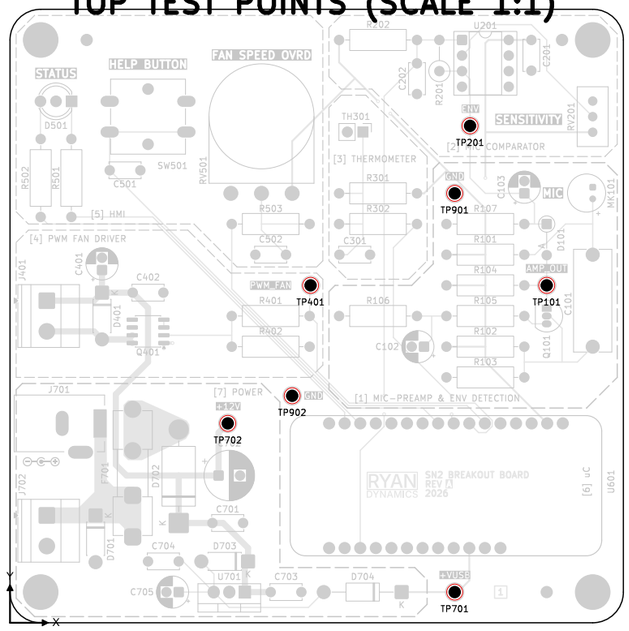
DRN 2024-04-13 R. HICKS			
CHK			
ENG APP	SN2 BREAKOUT BOARD		
MFR APP			
SHEET PATH	GIT HASH 9d7cd56	DRAWING No FABRICATION	A3
FILENAME Sensor-Node-2-Board.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased)	SHEET 4 OF 8

SN2 BREAKOUT BOARD FABRICATION DOCUMENT

TOP TEST POINTS (SCALE 1:1)



Ref.	Net	X [mm]	Y [mm]
TP101	AMP_OUT	87.50	55.00
TP201	ENV	75.00	81.00
TP401	PWM_FAN	49.00	55.00
TP701	+VUSB	72.50	5.00
TP702	+12V	35.50	32.50
TP901	GND	72.50	70.00
TP902	GND	46.00	37.00

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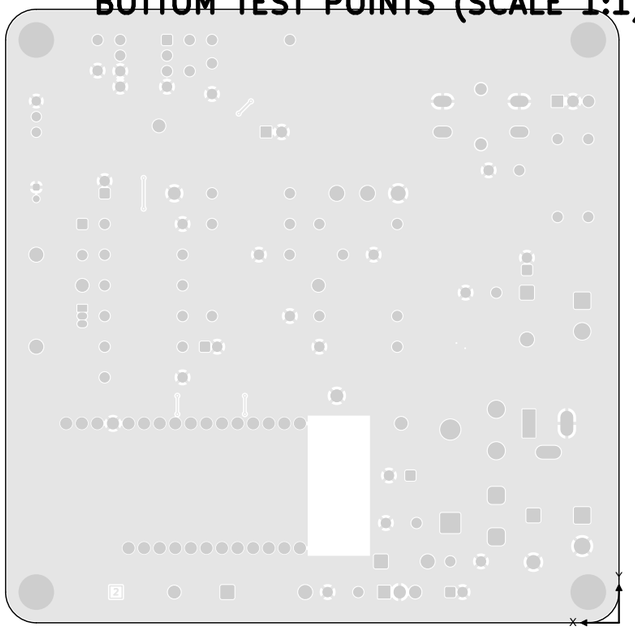
DRN 2024-04-13 R. HICKS		
CHK		
ENG APP	SN2 BREAKOUT BOARD	
MFR APP		
SHEET PATH	GIT HASH 9d7cd56	DRAWING No FABRICATION — A3
FILENAME Sensor-Node-2-Board.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased) — SHEET 5 OF 8

SN2 BREAKOUT BOARD FABRICATION DOCUMENT

BOTTOM TEST POINTS (SCALE 1:1)



Ref.	Net	X [mm]	Y [mm]

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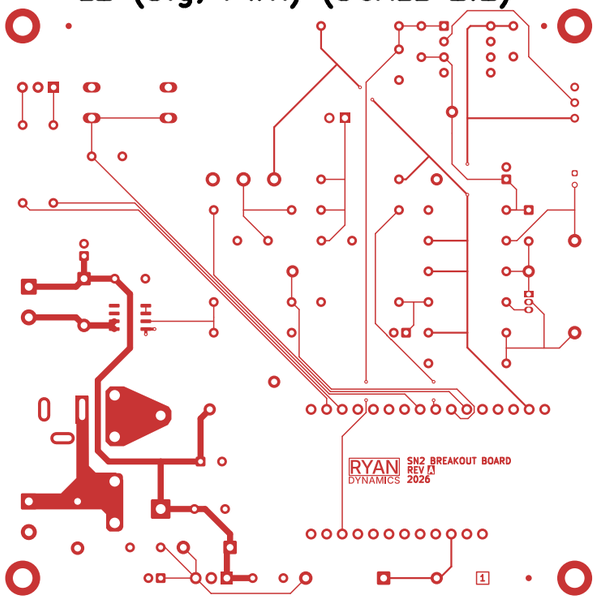
CHK			
ENG APP	SN2 BREAKOUT BOARD		
MFR APP			
SHEET PATH	GIT HASH 9d7cd56	DRAWING No FABRICATION	A3
FILENAME Sensor-Node-2-Board.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased)	SHEET 6 OF 8

SN2 BREAKOUT BOARD FABRICATION DOCUMENT

L1 (Sig, PWR) (SCALE 1:1)



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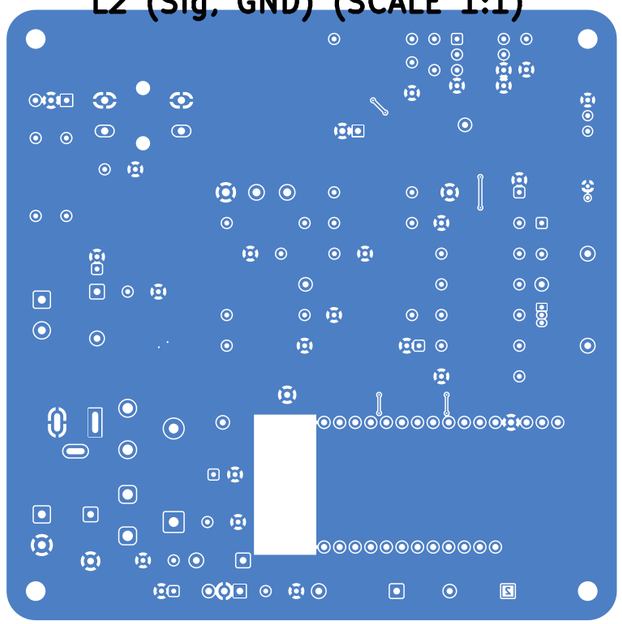
DRN 2024-04-13 R. HICKS			
CHK			
ENG APP	SN2 BREAKOUT BOARD		
MFR APP			
SHEET PATH	GIT HASH 9d7cd56	DRAWING No FABRICATION	A3
FILENAME Sensor-Node-2-Board.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased) SHEET 7 OF 8	

SN2 BREAKOUT BOARD FABRICATION DOCUMENT

L2 (Sig, GND) (SCALE 1:1)



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CHK			
ENG APP	SN2 BREAKOUT BOARD		
MFR APP			
SHEET PATH	GIT HASH 9d7cd56	DRAWING No FABRICATION	A3
FILENAME Sensor-Node-2-Board.kicad_pcb	VARIANT PRELIMINARY	REVISION + (Unreleased) SHEET 8 OF 8	